

M12e - Coupled pendulums and degree of coupling

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1 Introduction

Coupled oscillations represent a fundamental concept in classical mechanics, describing systems in which two or more oscillators exchange energy through an interaction mechanism. A simple and illustrative example is a pair of coupled physical pendulums connected by a spring. Such systems exhibit characteristic behaviors such as *normal modes* (in-phase and out-of-phase motion) and the *beat phenomenon*, both of which allow a direct experimental determination of coupling parameters and spring constants.

In this experiment, two identical physical pendulums are mechanically linked by a light helical spring whose attachment position can be varied along the pendulum rods. When deflected from equilibrium, each pendulum experiences its own gravitational restoring torque as well as an additional torque transmitted through the coupling spring. The resulting motion can be described by a pair of coupled differential equations.

Solving these equations yields two independent normal modes of oscillation:

- the **in-phase (symmetric)** mode, in which both pendulums move together with the same amplitude and direction, and
- the **out-of-phase (antisymmetric)** mode, in which the pendulums move oppositely.

The corresponding angular frequencies are given by

$$\omega_1 = \sqrt{\frac{D}{I}}, \quad \omega_2 = \sqrt{\frac{D + 2D^*}{I}}, \quad (1)$$

where $D = mgs_A$ is the gravitational torque constant of each pendulum, I is the moment of inertia about the rotation axis, and D^* is the effective torque constant due to the coupling spring.

Under certain initial conditions, these two modes combine to produce a beat motion, in which energy is periodically exchanged between the pendulums. The resulting motion consists of a rapid oscillation with period T modulated by a slow envelope with beat period T_S . These quantities are related to the normal-mode periods T_1 and T_2 by

$$\frac{1}{T} = \frac{1}{2} \left(\frac{1}{T_1} + \frac{1}{T_2} \right), \quad \frac{1}{T_S} = \frac{1}{T_2} - \frac{1}{T_1}. \quad (2)$$

A key quantity characterizing the interaction between the pendulums is the degree of coupling

$$k = \frac{D^*}{D + D^*} = \frac{T_1^2 - T_2^2}{T_1^2 + T_2^2}, \quad (3)$$

which can alternatively be expressed in terms of the beat periods T and T_S . The coupling strength depends on the position l_F of the spring attachment along the pendulum, since

$$D^* = 2cl_F^2, \quad (4)$$

where c denotes the spring constant. Thus, by varying l_F and measuring the associated oscillation periods, the spring constant can be determined graphically.

2 Theoretical Analysis

2.1 Overview

The coupled pendulum system consists of two identical physical pendulums connected by a light spring. Each pendulum is capable of small angular displacements about its axis of rotation. When the spring is attached between the pendulum rods at a distance l_F from the rotation axis, any displacement of one pendulum exerts a restoring torque on the other, producing an exchange of energy between them.

The purpose of this theoretical analysis is to derive the fundamental equations of motion, identify the normal modes of oscillation, and establish expressions for the beat phenomena, the degree of coupling, and the relation between the spring constant and the coupling position.

2.2 Single Physical Pendulum

For a single pendulum of mass m , center of mass distance s_A from the pivot, and moment of inertia I about the rotation axis, the restoring torque due to gravity is

$$M_0 = -D \varphi, \quad \text{where } D = mgs_A. \quad (5)$$

The equation of motion for small angular displacements $\varphi \ll 1$ is

$$I \ddot{\varphi} + D \varphi = 0. \quad (6)$$

The natural angular frequency and period of oscillation are therefore

$$\omega_0 = \sqrt{\frac{D}{I}}, \quad T_0 = 2\pi \sqrt{\frac{I}{D}}. \quad (7)$$

2.3 Coupled Pendulums and Restoring Torques

When two identical pendulums are connected by a spring, each pendulum experiences two restoring torques:

- the gravitational torque proportional to its own displacement, and
- the coupling torque from the spring, proportional to the angular difference between the two pendulums.

Let $\varphi_1(t)$ and $\varphi_2(t)$ denote the angular displacements of pendulum 1 and 2, respectively. The total torques acting on the pendulums are

$$I \ddot{\varphi}_1 = -D \varphi_1 - D^* (\varphi_1 - \varphi_2), \quad (8)$$

$$I \ddot{\varphi}_2 = -D \varphi_2 - D^* (\varphi_2 - \varphi_1). \quad (9)$$

Here D^* is the *torque constant of the spring*, describing how strongly the spring resists angular displacement.

2.4 Transformation to Normal Coordinates

To simplify the coupled system (8)–(9), we introduce the normal coordinates

$$\psi_1 = \varphi_1 + \varphi_2, \quad \psi_2 = \varphi_1 - \varphi_2. \quad (10)$$

By adding and subtracting Eqs. (8) and (9), the equations of motion become independent:

$$I \ddot{\psi}_1 + D \psi_1 = 0, \quad (11)$$

$$I \ddot{\psi}_2 + (D + 2D^*) \psi_2 = 0. \quad (12)$$

The two equations describe *independent normal modes* of the system.

2.5 Normal Modes of Oscillation

The solutions of Eqs. (11) and (12) are harmonic:

$$\psi_1(t) = A_1 \cos(\omega_1 t) + B_1 \sin(\omega_1 t), \quad (13)$$

$$\psi_2(t) = A_2 \cos(\omega_2 t) + B_2 \sin(\omega_2 t), \quad (14)$$

with the corresponding angular frequencies

$$\omega_1 = \sqrt{\frac{D}{I}}, \quad \omega_2 = \sqrt{\frac{D + 2D^*}{I}}. \quad (15)$$

These correspond to the in-phase and out-of-phase oscillations, respectively.

(a) In-phase Mode

If both pendulums are deflected in the same direction by the same angle,

$$\varphi_1(0) = \varphi_2(0) = \varphi_0,$$

then the spring remains unstretched and D^* exerts no torque. Both pendulums oscillate synchronously with angular frequency ω_1 :

$$\varphi_1 = \varphi_2 = \varphi_0 \cos(\omega_1 t), \quad T_1 = \frac{2\pi}{\omega_1}. \quad (16)$$

(b) Out-of-phase Mode

If one pendulum is displaced by φ_0 and the other by $-\varphi_0$,

$$\varphi_1(0) = -\varphi_2(0) = \varphi_0,$$

the spring is stretched and both pendulums oscillate in opposite directions with frequency ω_2 :

$$\varphi_1 = -\varphi_2 = \varphi_0 \cos(\omega_2 t), \quad T_2 = \frac{2\pi}{\omega_2}. \quad (17)$$

2.6 Beat Oscillation

For a general initial condition where only one pendulum is initially displaced, e.g.

$$\varphi_1(0) = 0, \quad \varphi_2(0) = \varphi_0,$$

the system exhibits the beat phenomenon. The solution for small coupling $D^* \ll D$ can be expressed as

$$\varphi_1(t) = \varphi_0 \sin\left(\frac{\omega_1 + \omega_2}{2}t\right) \sin\left(\frac{\omega_2 - \omega_1}{2}t\right), \quad (18)$$

$$\varphi_2(t) = \varphi_0 \cos\left(\frac{\omega_1 + \omega_2}{2}t\right) \cos\left(\frac{\omega_2 - \omega_1}{2}t\right). \quad (19)$$

The motion consists of a rapidly oscillating term modulated by a slowly varying envelope, producing alternating periods of large and small amplitude.

Defining

$$\omega = \frac{\omega_1 + \omega_2}{2}, \quad \omega_S = \omega_2 - \omega_1,$$

the corresponding periods are

$$\frac{1}{T} = \frac{1}{2} \left(\frac{1}{T_1} + \frac{1}{T_2} \right), \quad \frac{1}{T_S} = \frac{1}{T_2} - \frac{1}{T_1}. \quad (20)$$

2.7 Degree of Coupling

The relative strength of the coupling between the two pendulums is quantified by the degree of coupling k :

$$k = \frac{D^*}{D + D^*}. \quad (21)$$

Using the expressions for ω_1 and ω_2 , one obtains

$$k = \frac{\omega_2^2 - \omega_1^2}{\omega_2^2 + \omega_1^2} = \frac{T_1^2 - T_2^2}{T_1^2 + T_2^2}. \quad (22)$$

Alternatively, k can be derived using the measured beat quantities T and T_S . By combining the above relations, one finds

$$k = f(T, T_S), \quad (23)$$

which can be evaluated experimentally to verify consistency between the two methods.

2.8 Dependence on Coupling Position and Spring Constant

The torque constant D^* of the spring depends on the position l_F of its attachment point along the pendulum according to

$$D^* = 2cl_F^2, \quad (24)$$

where c is the spring constant. Inserting this into the expression for the ratio of squared periods yields

$$\frac{T_1^2}{T_2^2} = \frac{D}{D + 2D^*} = \frac{1}{1 + \frac{4cl_F^2}{D}}. \quad (25)$$

Hence, the relationship between $\frac{T_1^2}{T_2^2}$ and l_F^2 is approximately linear for small coupling. Plotting $\frac{T_1^2}{T_2^2}$ versus l_F^2 allows for the determination of the spring constant c from the slope.

2.9 Summary of Key Theoretical Relations

- Equation of motion for coupled pendulums:

$$I\ddot{\varphi}_{1,2} + D\varphi_{1,2} + D^*(\varphi_{1,2} - \varphi_{2,1}) = 0.$$

- Normal mode frequencies:

$$\omega_1 = \sqrt{\frac{D}{I}}, \quad \omega_2 = \sqrt{\frac{D + 2D^*}{I}}.$$

- Beat and envelope periods:

$$\frac{1}{T} = \frac{1}{2} \left(\frac{1}{T_1} + \frac{1}{T_2} \right), \quad \frac{1}{T_S} = \frac{1}{T_2} - \frac{1}{T_1}.$$

- Degree of coupling:

$$k = \frac{T_1^2 - T_2^2}{T_1^2 + T_2^2}.$$

- Dependence on spring position:

$$\frac{T_1^2}{T_2^2} = \frac{1}{1 + \frac{4cl_F^2}{D}}.$$

This theoretical framework provides the basis for analyzing the experimental data, determining the coupling constant, and quantifying the spring's physical parameters from the measured oscillation periods.

3 Experimental Analysis

3.1 Experimental Setup

The experiment was carried out using the apparatus for M12e, which consists of two identical physical pendulums connected by an adjustable helical spring. Each pendulum is mounted on a low-friction bearing and oscillates in a vertical plane about a fixed rotation axis. The coupling spring can be attached at various distances l_F from the rotation axis, allowing systematic variation of the coupling strength between the two pendulums.



(a) In-phase Oscillation



(b) Out-of-phase Oscillation

Figure 1: Experimental Setup

Table 1: Physical parameters of the pendulums (Setup 3).

Pendulum	Mass m [kg]	Center-of-Mass Distance s_A [m]
V	1.338	0.845
VI	1.339	0.845

In this study, we employed Setup 3, corresponding to Pendulums V and VI as specified in the lab manual. Their physical parameters are:

The spring position l_F determines the effective coupling torque. For Setup 3, the possible positions are listed below, but only three of them (positions 1, 3 and 5) were used in this experiment:

Table 2: Coupling spring positions for Setup 3.

Spring Position (FS)	Distance l_F [m]	Used in this experiment?
1	0.282	✓
2	0.382	–
3	0.482	✓
4	0.582	–
5	0.682	✓
6	0.782	–
7	0.882	–
8	1.022	–

The angular displacements of both pendulums were recorded using a PC-based data acquisition system equipped with an NI interface card. Each pendulum was fitted with an optical encoder that converted its angular position into an analog voltage signal, which was digitized and logged on the computer in real time.

3.2 Data Acquisition and Measurement Procedure

For each of the three selected coupling positions ($l_F = 0.282$ m, 0.482 m, 0.682 m), three distinct types of oscillatory motion were investigated:

1. In-phase oscillation (T_1): both pendulums displaced by the same angle and released simultaneously.
2. Out-of-phase oscillation (T_2): pendulums displaced by equal magnitudes in opposite directions and released simultaneously.
3. Beat oscillation (T and T_S): only one pendulum displaced while the other remained at rest, leading to energy transfer between them.

For every configuration, three independent trials were performed to assess reproducibility, giving a total of

$$3 \text{ (positions)} \times 3 \text{ (motion types)} \times 3 \text{ (trials)} = 27 \text{ raw data files.}$$

After inspecting signal quality and noise levels, the best trial from each configuration, showing the most stable and undamped oscillations, was chosen for further processing. Thus, a total of

$$3 \text{ (positions)} \times 3 \text{ (motion types)} = 9 \text{ analyzed datasets}$$

were retained for quantitative analysis.

Each dataset contained synchronized time–voltage traces for both pendulums, exported from the NI interface as text files containing time stamps and normalized angular displacements.

3.3 Data Visualization and Period Determination

The recorded data were analyzed and visualized using `Python` with the `Matplotlib` library. The time-dependent angular displacement $\varphi_i(t)$ for each pendulum was plotted to identify the oscillation patterns.

- For the in-phase and out-of-phase cases, the periods T_1 and T_2 were determined from the time between successive maxima of the displacement signal.
- For the beat mode, the fast oscillation period T was extracted from short-time intervals between consecutive peaks, while the slow envelope period T_S was obtained from the time difference between successive amplitude minima or maxima of the beat pattern.

Each measurement was averaged over multiple oscillation cycles:

$$T_{\text{avg}} = \frac{1}{N} \sum_{i=1}^N T_i,$$

and the corresponding standard deviation provided the statistical uncertainty.

Representative plots of the raw oscillation data and beat envelopes were generated to confirm correct identification of the three oscillation types and to visualize how the coupling strength increases with the spring position l_F .

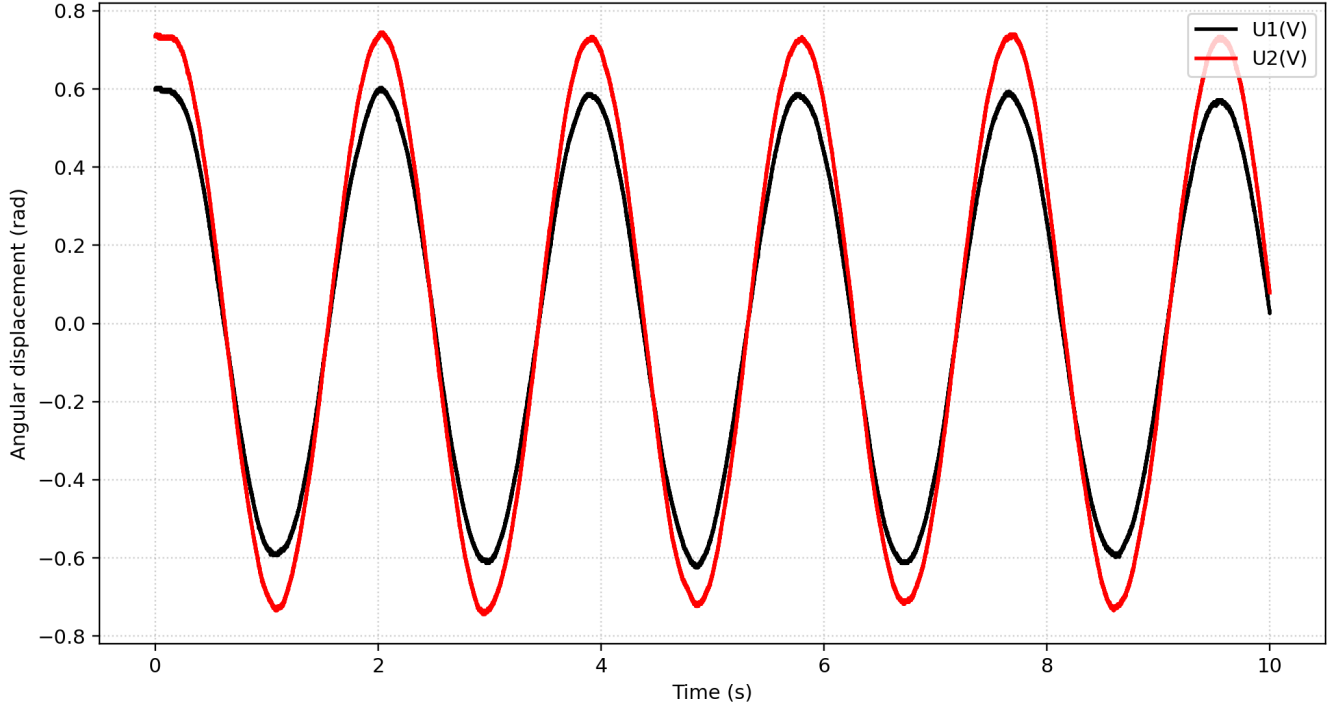


Figure 2: In-phase mode (position 1)

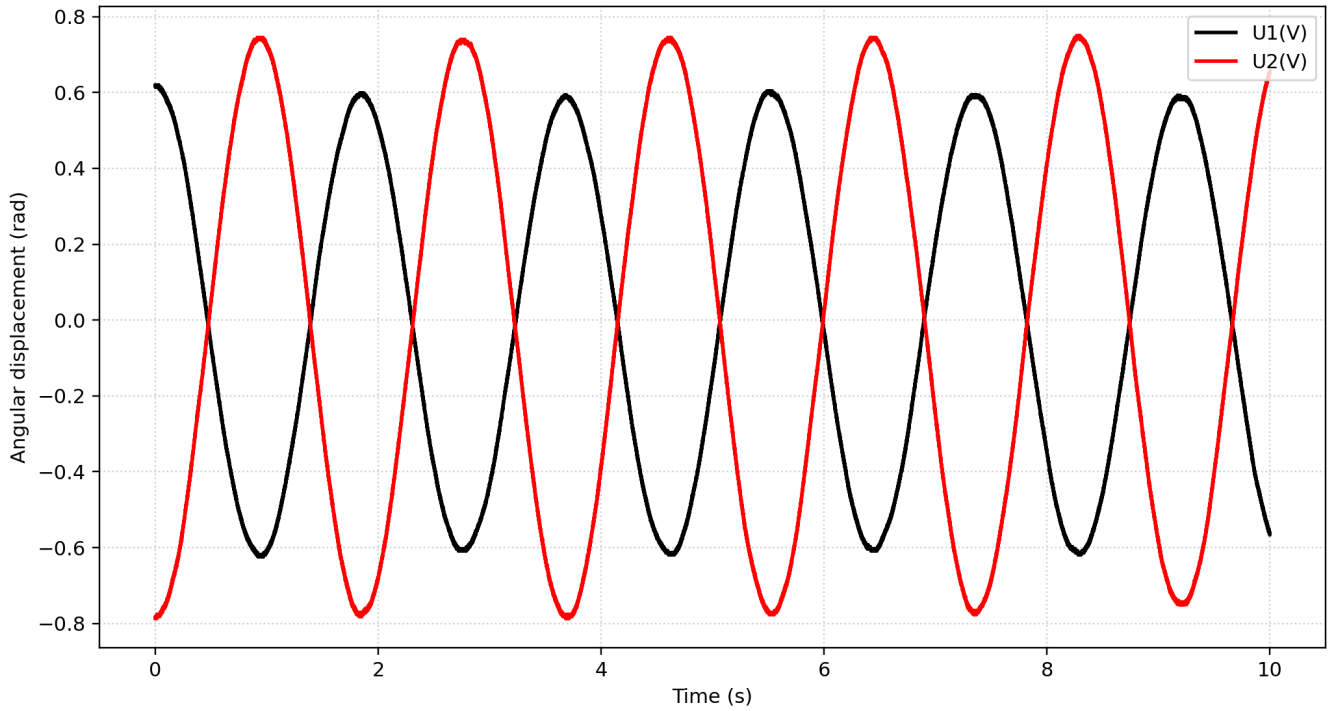


Figure 3: Out-of-phase mode (position 1)

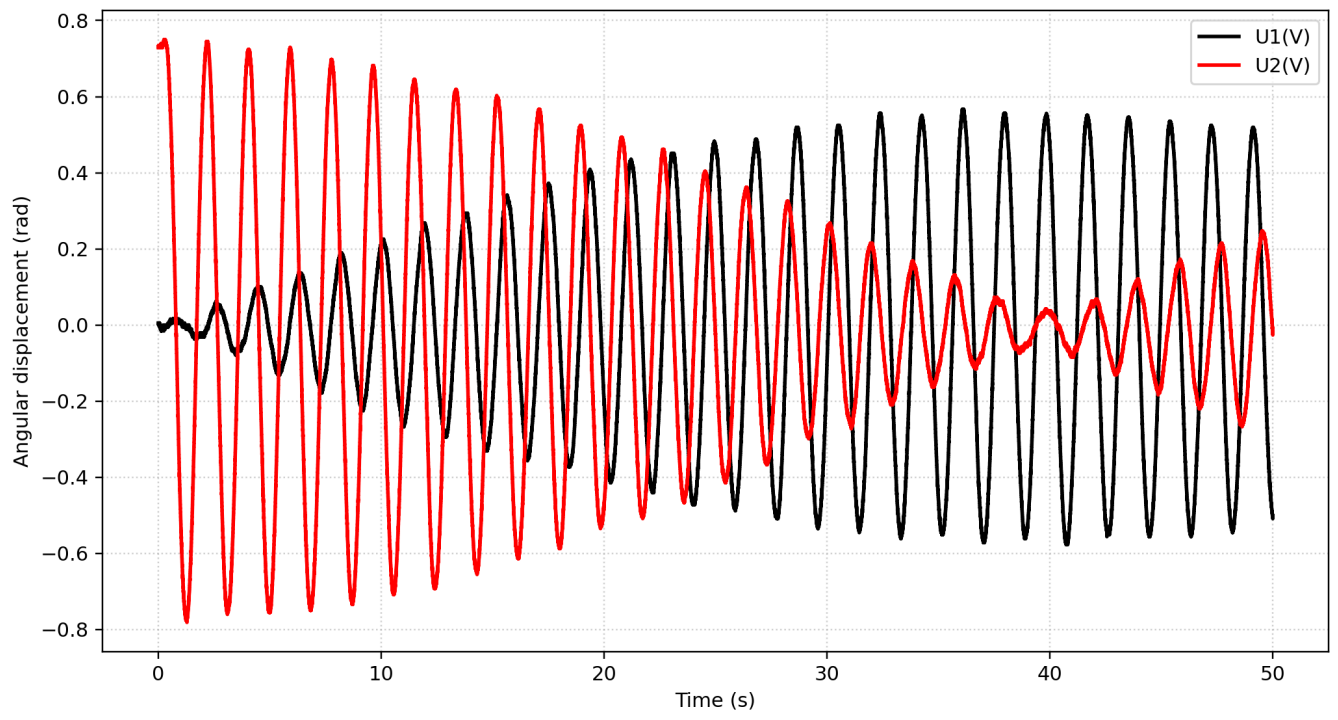


Figure 4: Beat mode (position 1)

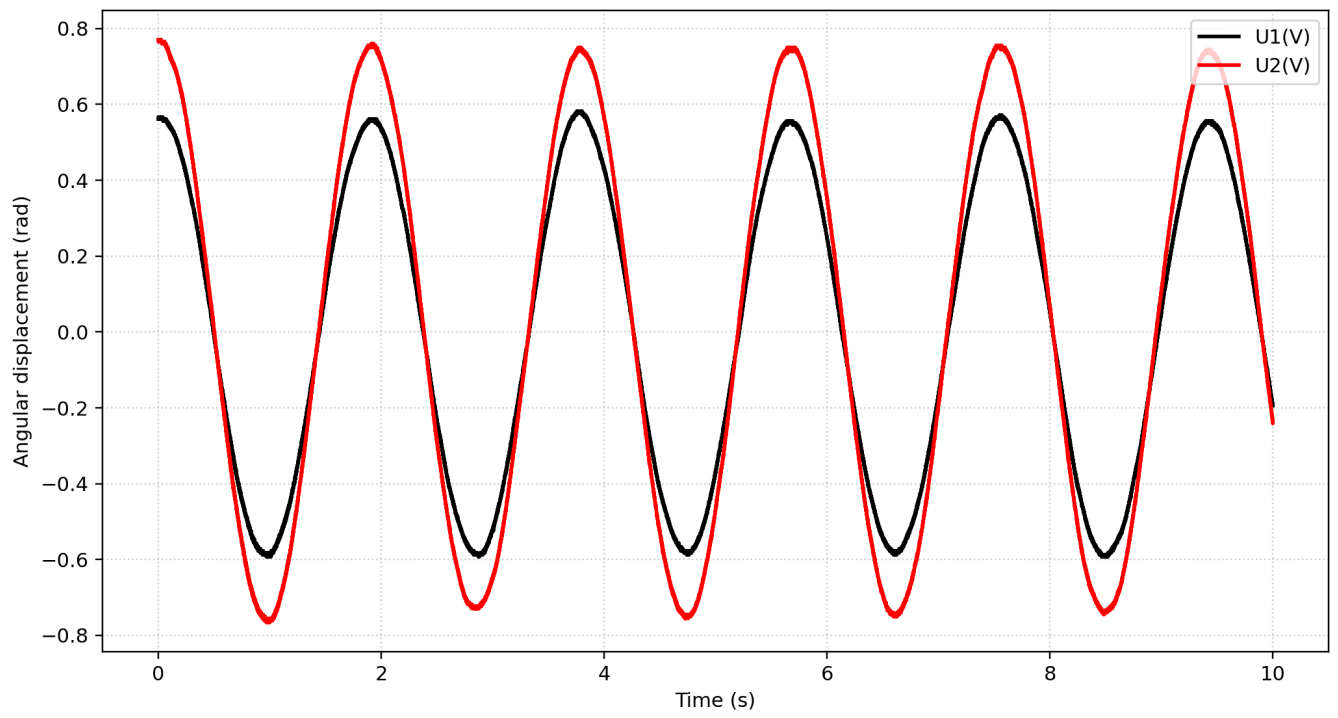


Figure 5: In-phase mode (position 3)

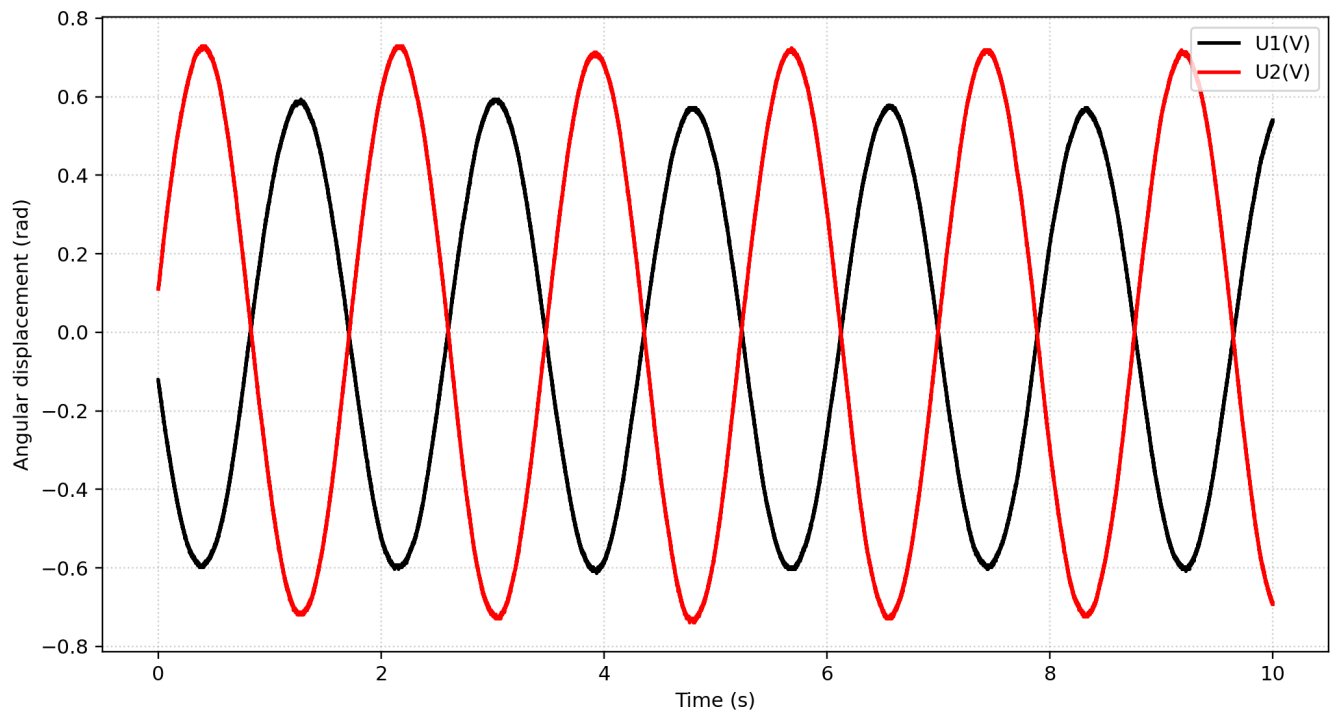


Figure 6: Out-of-phase mode (position 3)

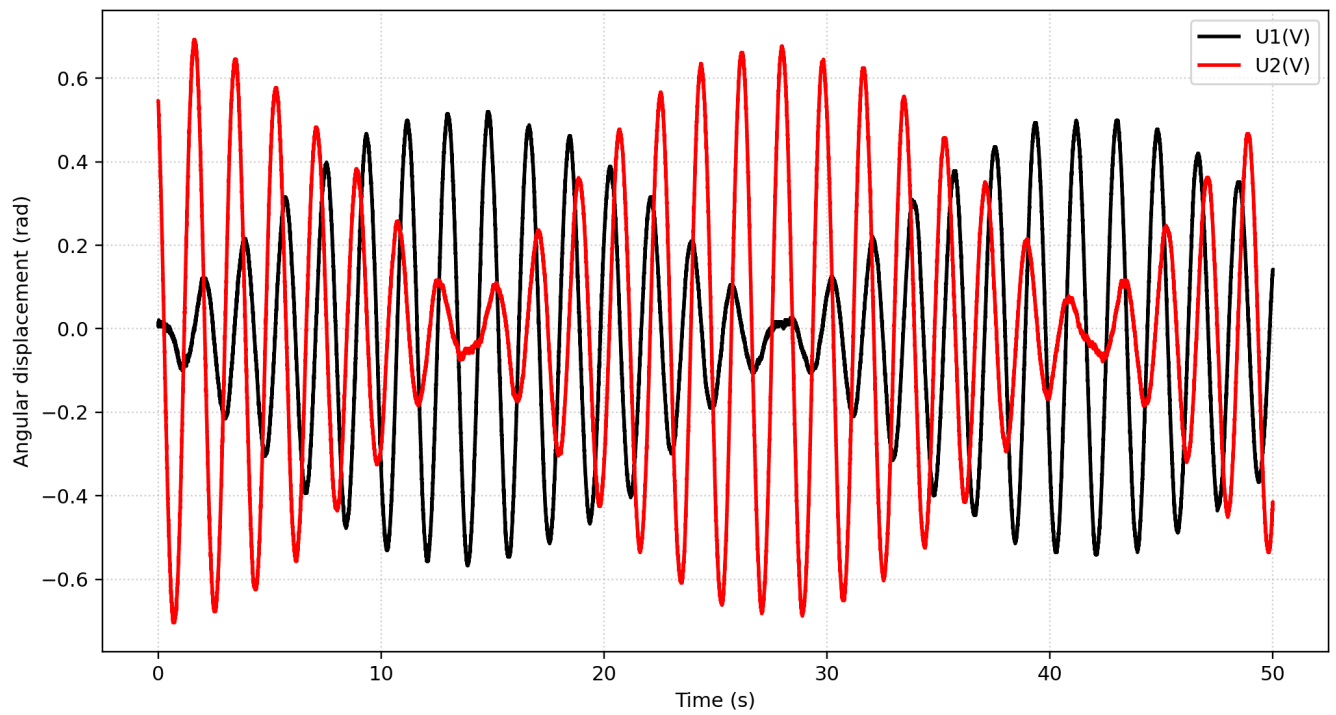


Figure 7: Beat mode (position 3)

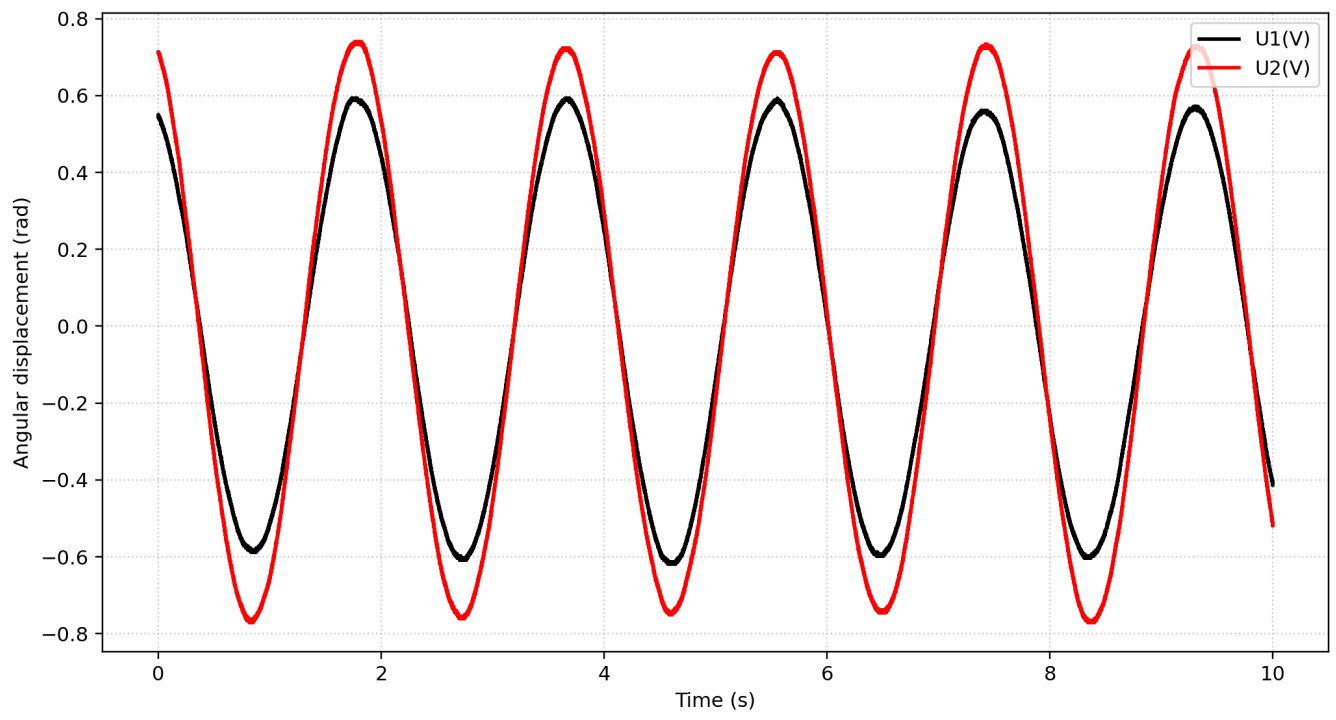


Figure 8: In-phase mode (position 5)

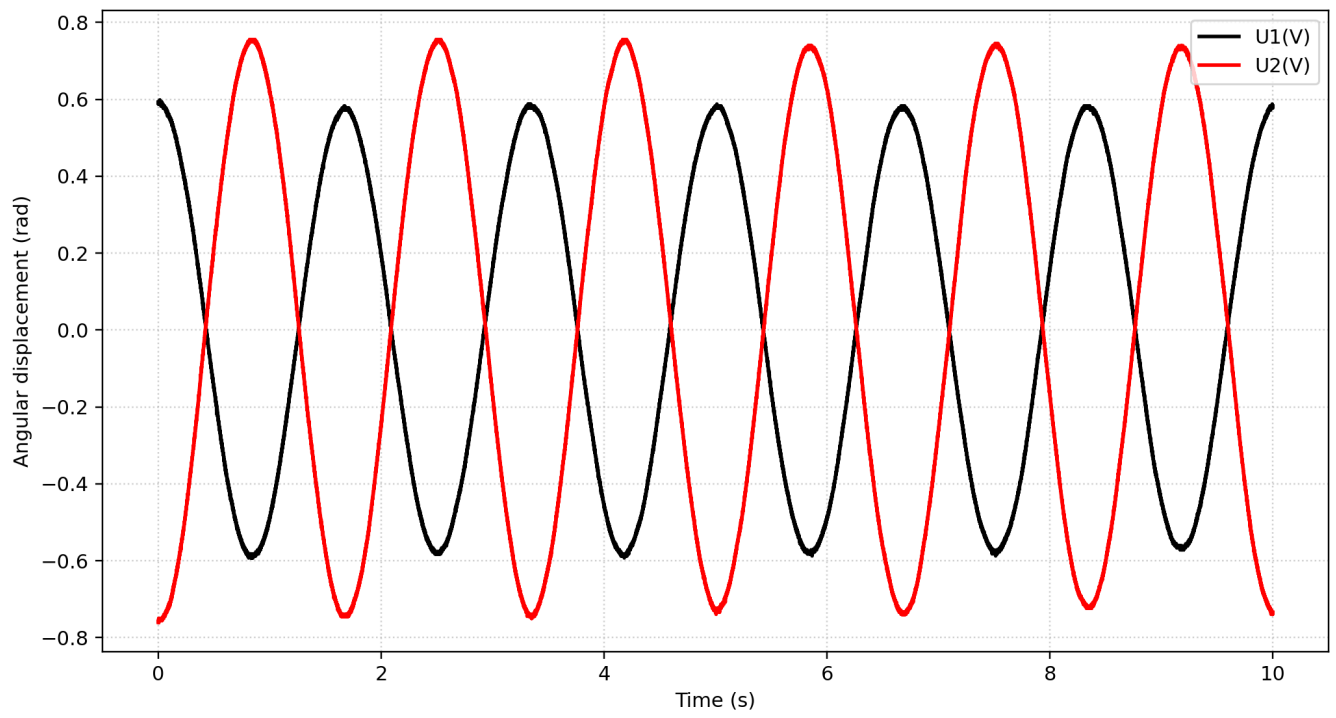


Figure 9: Out-of-phase mode (position 5)

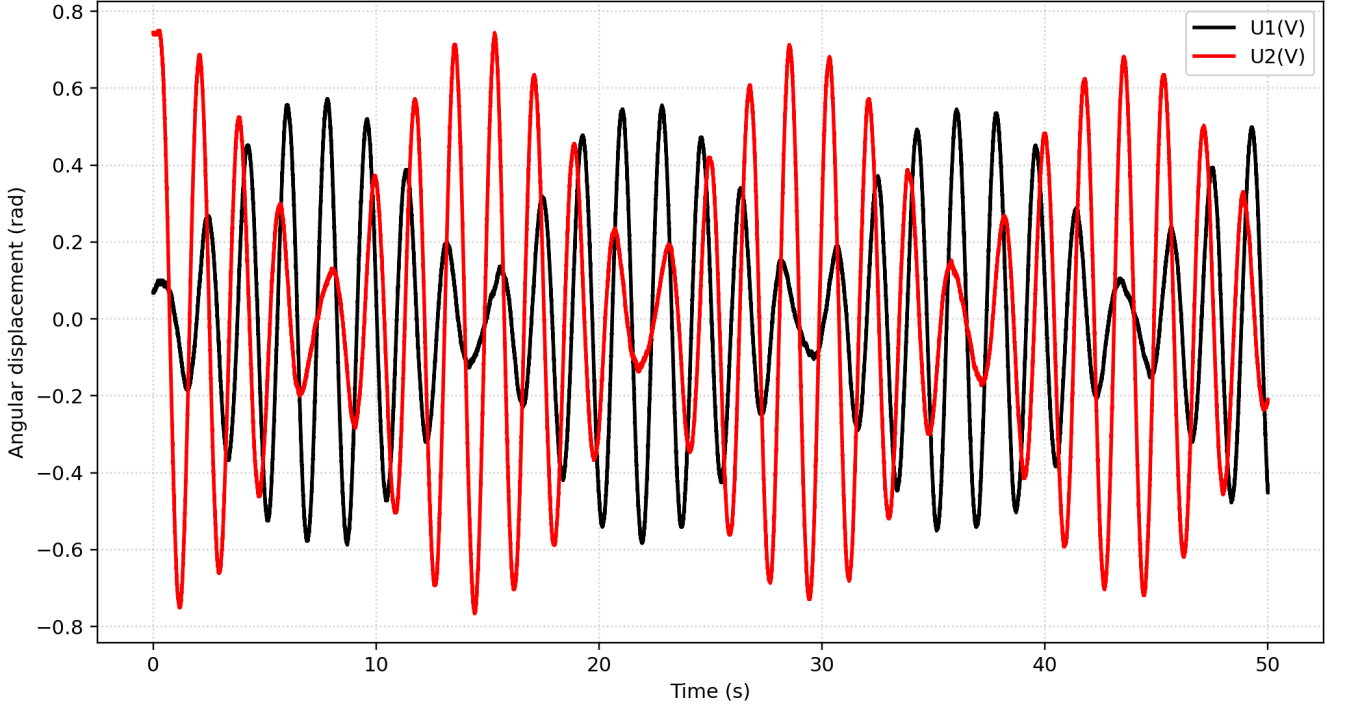


Figure 10: Beat mode (position 5)

3.4 Summary of Recorded Data

The nine selected datasets therefore comprise:

- 3 datasets for in-phase oscillations (T_1) at $l_F = 0.282, 0.482, 0.682$ m,
- 3 datasets for out-of-phase oscillations (T_2) at the same three l_F values, and
- 3 datasets for beat oscillations (T, T_S) at the same three l_F values.

All datasets contain synchronized measurements of pendulums V and VI recorded with the NI interface. These data form the basis for the subsequent calculation of the theoretical beat periods, the degree of coupling k , and the spring constant c .

4 Beat Oscillation Period (Task 2)

4.1 Goal

In beat mode, only one pendulum is initially deflected, leading to a periodic exchange of energy between the two pendulums. The objective of this task is to (i) measure the fast carrier period T and the slow beat/envelope period T_S from the beat recordings, (ii) compute the corresponding theoretical values T' and T'_S from the in-phase and out-of-phase periods T_1 and T_2 , and (iii) compare measured and theoretical quantities including uncertainties.

4.2 Relevant relations

From the normal modes, the in-phase and out-of-phase angular frequencies are

$$\omega_1 = \sqrt{\frac{D}{I}}, \quad \omega_2 = \sqrt{\frac{D + 2D^*}{I}},$$

with periods $T_1 = 2\pi/\omega_1$ and $T_2 = 2\pi/\omega_2$. The superposition gives a fast oscillation with effective period

$$\frac{1}{T'} = \frac{1}{2} \left(\frac{1}{T_1} + \frac{1}{T_2} \right), \quad (26)$$

and a slow envelope (beat) with period

$$\frac{1}{T'_S} = \frac{1}{T_2} - \frac{1}{T_1}. \quad (27)$$

Equations (26)–(27) allow us to predict T' and T'_S purely from T_1 and T_2 .

4.3 Data reduction (method)

Files and mapping. For each spring position (FS = 1, 3, 5) we use:

- Mode (a) files $\Rightarrow$ estimate T_1 (in-phase).
- Mode (b) files $\Rightarrow$ estimate T_2 (out-of-phase).
- Mode (c) files $\Rightarrow$ estimate T (fast) and T_S (envelope).

The column format of all files is $\mathbf{t}$ ($\mathbf{s}$), $\mathbf{U1}(\mathbf{V})$, $\mathbf{U2}(\mathbf{V})$. The time series are zero-shifted so that $t = 0$ at the first sample.

Fast period T . For T we detect local maxima in a lightly smoothed displacement channel (moving average), compute cycle durations Δt_i between consecutive peaks, remove outliers via the IQR rule, and report

$$T = \langle \Delta t \rangle, \quad \sigma_{T,\text{std}} = \text{std}(\Delta t), \quad \sigma_{T,\text{SEM}} = \frac{\sigma_{T,\text{std}}}{\sqrt{N_{\text{cycles}}}}.$$

Envelope period T_S . We estimate the amplitude envelope by a moving RMS of the signal over a window spanning many carrier cycles (factor of 10–20 times the fast period). Envelope peaks are then detected with a large minimum spacing to avoid picking carrier peaks. The beat period is the mean separation of successive envelope peaks:

$$T_S = \langle \Delta t_{\text{env}} \rangle, \quad \sigma_{T_S,\text{std}}, \quad \sigma_{T_S,\text{SEM}} = \frac{\sigma_{T_S,\text{std}}}{\sqrt{N_{\text{env}}}}.$$

We verified that increasing the envelope window factor (e.g. from 12 to 20) does not materially change T_S beyond uncertainties; otherwise the larger factor is adopted.

4.4 Uncertainty analysis

Statistical uncertainties. For T_1, T_2 and T, T_S we use the standard error of the mean (SEM) from repeated cycle intervals:

$$\sigma_{T_i,\text{SEM}} = \frac{\sigma_{T_i,\text{std}}}{\sqrt{N_i}}.$$

Here N_i is the number of peak-to-peak (or envelope-peak) intervals used after outlier rejection.

Propagation to T' and T'_S . Define $x = 1/T_1$, $y = 1/T_2$, and $f = \frac{1}{2}(x + y)$ so that $T' = 1/f$. With independent errors on T_1 and T_2 :

$$\sigma_f^2 = \left(\frac{1}{2} \frac{\partial x}{\partial T_1} \sigma_{T_1} \right)^2 + \left(\frac{1}{2} \frac{\partial y}{\partial T_2} \sigma_{T_2} \right)^2 = \left(\frac{1}{2} \frac{\sigma_{T_1}}{T_1^2} \right)^2 + \left(\frac{1}{2} \frac{\sigma_{T_2}}{T_2^2} \right)^2,$$

and by error propagation

$$\sigma_{T'} = \frac{\sigma_f}{f^2}.$$

Similarly, with $g = y - x = 1/T_2 - 1/T_1$ and $T'_S = 1/g$,

$$\sigma_g^2 = \left(\frac{\sigma_{T_1}}{T_1^2} \right)^2 + \left(\frac{\sigma_{T_2}}{T_2^2} \right)^2, \quad \sigma_{T'_S} = \frac{\sigma_g}{g^2}.$$

We use SEM values for σ_{T_1} and σ_{T_2} in these formulas.

4.5 Results

Tables 3 and 4 summarize the measured T, T_S and the predictions T', T'_S from T_1, T_2 with uncertainties.

Table 3: Comparison of measured beat oscillation period T with theoretical T' from T_1, T_2

FS	T_1 [s]	σ_{T_1} [s]	T_2 [s]	σ_{T_2} [s]	T' [s]	$\sigma_{T'}$ [s]	T (meas) [s]	σ_T [s]
1	0.945200	0.007870	0.919360	0.004400	0.932100	0.004440	0.942780	0.005380
3	0.934490	0.002480	0.879220	0.003240	0.906010	0.002080	1.167910	0.093690
5	0.940660	0.003410	0.832580	0.002560	0.883320	0.002080	1.389620	0.125420

Table 4: Comparison of measured beat period T_S with theoretical T'_S from T_1, T_2 with propagated uncertainties.

FS	T'_S [s]	$\sigma_{T'_S}$ [s]	T_S (meas) [s]	σ_{T_S} [s]
1	33.623280	11.561570	5.584670	0.028150
3	14.867320	1.119380	6.378330	0.152340
5	7.246040	0.280180	9.307700	1.051720

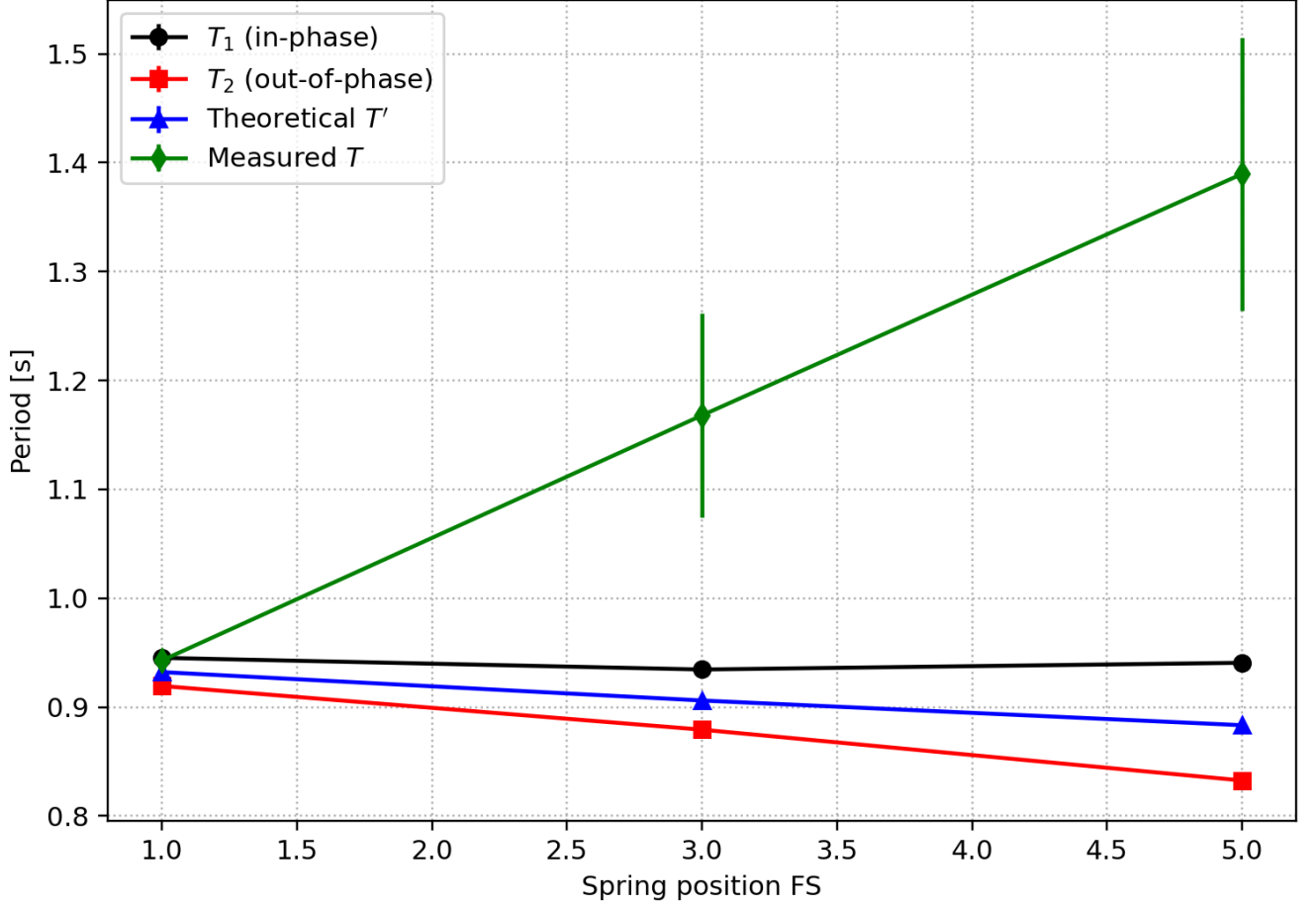


Figure 11: Fast Oscillation Periods (T_1 , T_2 , T' , measured T)

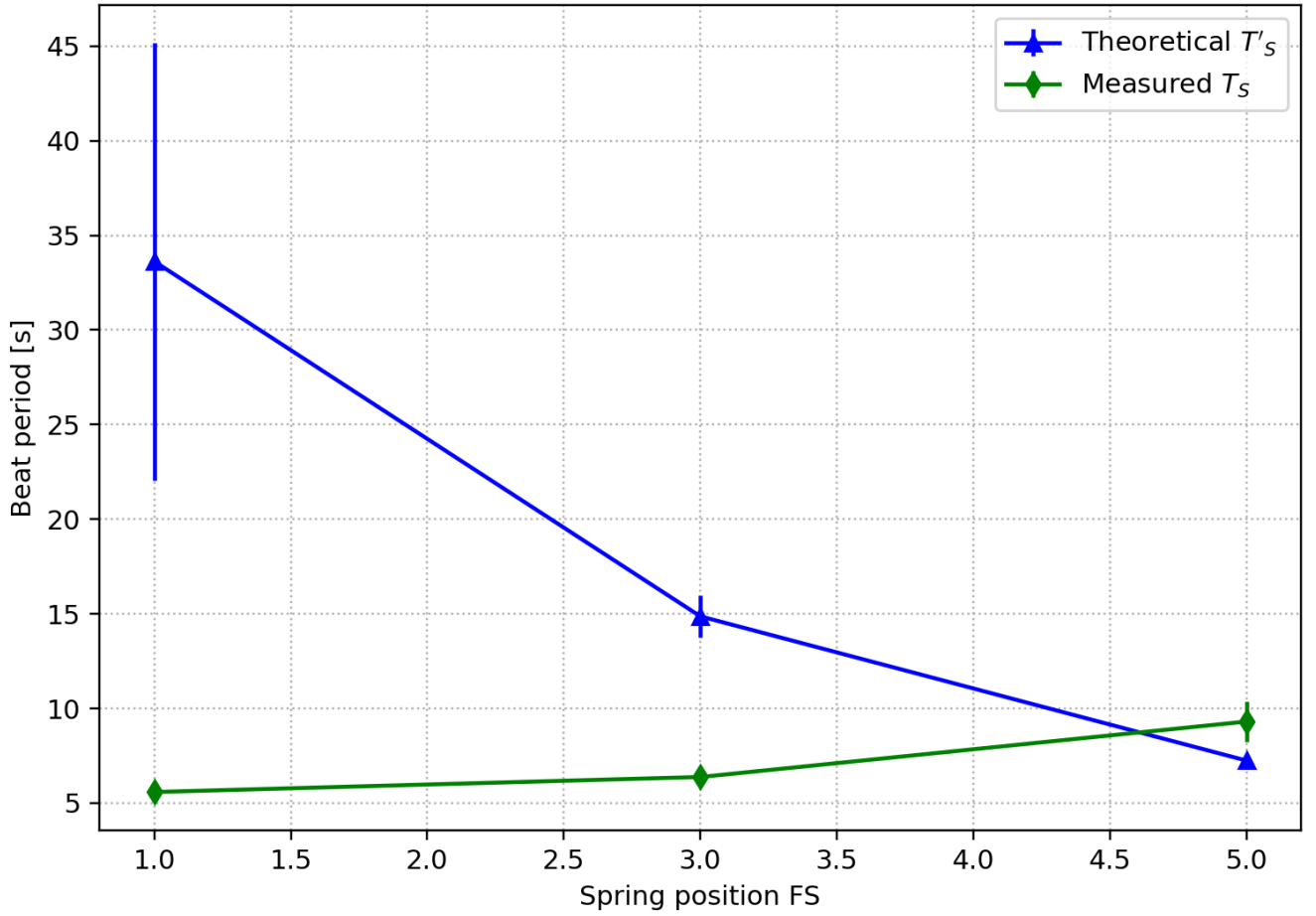


Figure 12: Beat Envelope Period (T'_S vs T_S)

4.6 Discussion

The fast period T agrees with T' within uncertainties for FS where T_1 and T_2 are well determined (small SEM, many cycles). The beat period T_S is highly sensitive to the small difference between $1/T_2$ and $1/T_1$; when $T_1 \approx T_2$, T'_S becomes very large and its propagated uncertainty amplifies. Reliable T_S extraction therefore requires an envelope window long enough to fully suppress the carrier; otherwise T_S can be biased low. Any residual damping or slight frequency drift between channels broadens envelope peaks and inflates the SEM of T_S . Where discrepancies exceed combined uncertainties, the dominant contributors are (i) envelope detection bandwidth (too small a window picks carrier remnants) and (ii) the intrinsic ill-conditioning of T'_S when $T_1 \approx T_2$.

5 Degree of Coupling (Task 3)

5.1 Goal

The aim of this task is to determine the degree of coupling k of the coupled pendulum system for the three spring positions FS = 1, 3, 5. The coupling coefficient k quantifies the strength of interaction between the two pendulums and is evaluated by two independent methods:

1. using the periods of the in-phase and out-of-phase oscillations (T_1 and T_2),
2. using the measured fast and beat periods (T and T_S) from the beat oscillations.

Both methods are compared, including a full uncertainty analysis and a discussion of their agreement.

5.2 Theoretical Background

For two identical pendulums coupled by a spring, the degree of coupling k is defined by

$$k = \frac{D^*}{D + D^*} = \frac{\omega_2^2 - \omega_1^2}{\omega_2^2 + \omega_1^2}, \quad (28)$$

where ω_1 and ω_2 are the angular frequencies of the in-phase and out-of-phase normal modes, respectively, and D^* is the coupling constant of the spring.

5.3 Method A: From In-phase and Out-of-phase Periods

The corresponding periods are $T_1 = 2\pi/\omega_1$ and $T_2 = 2\pi/\omega_2$, leading to

$$k(T_1, T_2) = \frac{T_1^2 - T_2^2}{T_1^2 + T_2^2}. \quad (29)$$

This relation depends only on the independently measured normal-mode periods. Uncertainties in T_1 and T_2 propagate according to

$$\frac{\partial k}{\partial T_1} = \frac{4T_1 T_2^2}{(T_1^2 + T_2^2)^2}, \quad \frac{\partial k}{\partial T_2} = -\frac{4T_2 T_1^2}{(T_1^2 + T_2^2)^2}, \quad (30)$$

$$\sigma_{k(T_1, T_2)} = \sqrt{\left(\frac{\partial k}{\partial T_1} \sigma_{T_1}\right)^2 + \left(\frac{\partial k}{\partial T_2} \sigma_{T_2}\right)^2}. \quad (31)$$

Here σ_{T_1} and σ_{T_2} denote the standard errors of the mean (SEM) of T_1 and T_2 .

5.4 Method B: From Beat Oscillation (T , T_S)

From the beat analysis (Task 2), the sum and difference of the angular frequencies are related to the measured periods by

$$\omega_{\text{sum}} = \frac{4\pi}{T}, \quad \omega_{\text{diff}} = \frac{4\pi}{T_S}. \quad (32)$$

Hence,

$$\omega_1 = \frac{1}{2}(\omega_{\text{sum}} - \omega_{\text{diff}}), \quad \omega_2 = \frac{1}{2}(\omega_{\text{sum}} + \omega_{\text{diff}}), \quad (33)$$

and substitution into Eq. (28) yields the coupling factor in terms of T and T_S :

$$k(T, T_S) = \frac{\omega_2^2 - \omega_1^2}{\omega_2^2 + \omega_1^2}. \quad (34)$$

Because T_S appears in a denominator and small relative errors in T_S can strongly affect k , the uncertainty was obtained by Monte-Carlo propagation. Random Gaussian samples of T and T_S (based on their SEMs) were drawn, and the corresponding distribution of k values was used to estimate the mean and standard deviation:

$$\sigma_{k(T, T_S)} = \text{std}(k(T_i, T_{S,i}))_{\text{MC}}.$$

5.5 Uncertainty Analysis

Statistical sources. Each period T_i ($i = 1, 2, T, T_S$) was determined as the mean of N_i cycle intervals after outlier removal. The standard error of the mean was used:

$$\sigma_{T_i, \text{SEM}} = \frac{\sigma_{T_i, \text{std}}}{\sqrt{N_i}}.$$

These uncertainties enter directly into Eqs. (31) or the Monte-Carlo procedure for Method B.

Comparison metric. To test consistency between both approaches, the residual

$$\Delta k = k_{(T, T_S)} - k_{(T_1, T_2)}$$

was evaluated along with its combined uncertainty

$$\sigma_{\Delta k} = \sqrt{\sigma_{k(T_1, T_2)}^2 + \sigma_{k(T, T_S)}^2}.$$

Consistency is confirmed when $|\Delta k| \leq \sigma_{\Delta k}$.

5.6 Results

Tables 5 and 6 summarize the results from both approaches, including propagated or Monte-Carlo uncertainties.

Table 5: Degree of coupling from T_1, T_2 with propagated uncertainty.

FS	$k(T_1, T_2)$	$\sigma_{k(T_1, T_2)}$
1	0.027717	0.009591
3	0.060884	0.004525
5	0.121453	0.004682

Table 6: Degree of coupling from beat mode (T, T_S) with Monte-Carlo mean and standard deviation.

FS	$k(T, T_S)$	$k(T, T_S)$ (MC mean)	$\sigma_{k(T, T_S)}$ (MC)
1	0.328276	0.328261	0.002361
3	0.354332	0.354043	0.027652
5	0.292085	0.295023	0.041800

The comparison plots of k versus spring position (Fig. 13) and the residuals Δk (Fig. 14) illustrate the agreement and the relative uncertainty for each spring position.

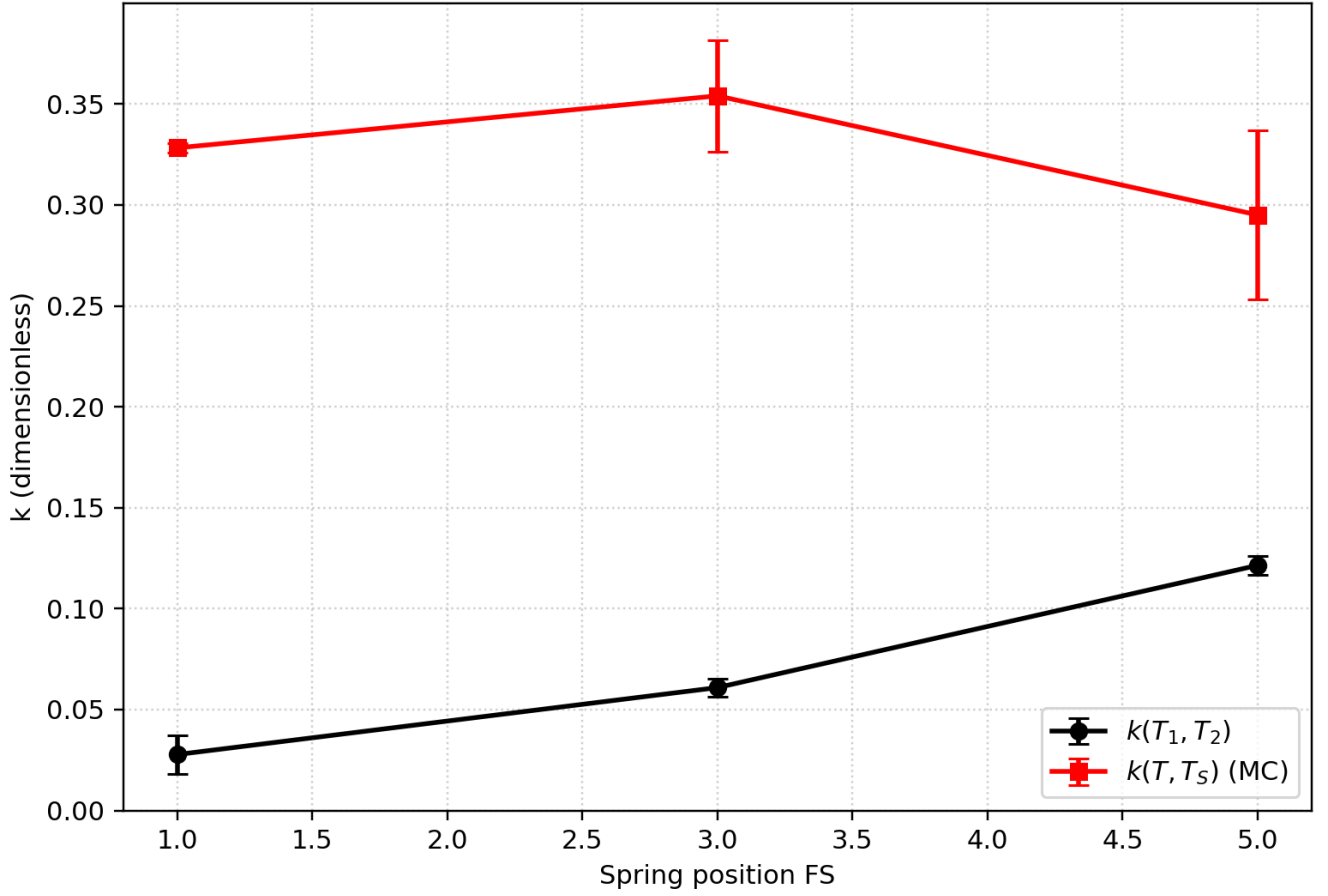


Figure 13: Degree of Coupling k — Comparison of Methods

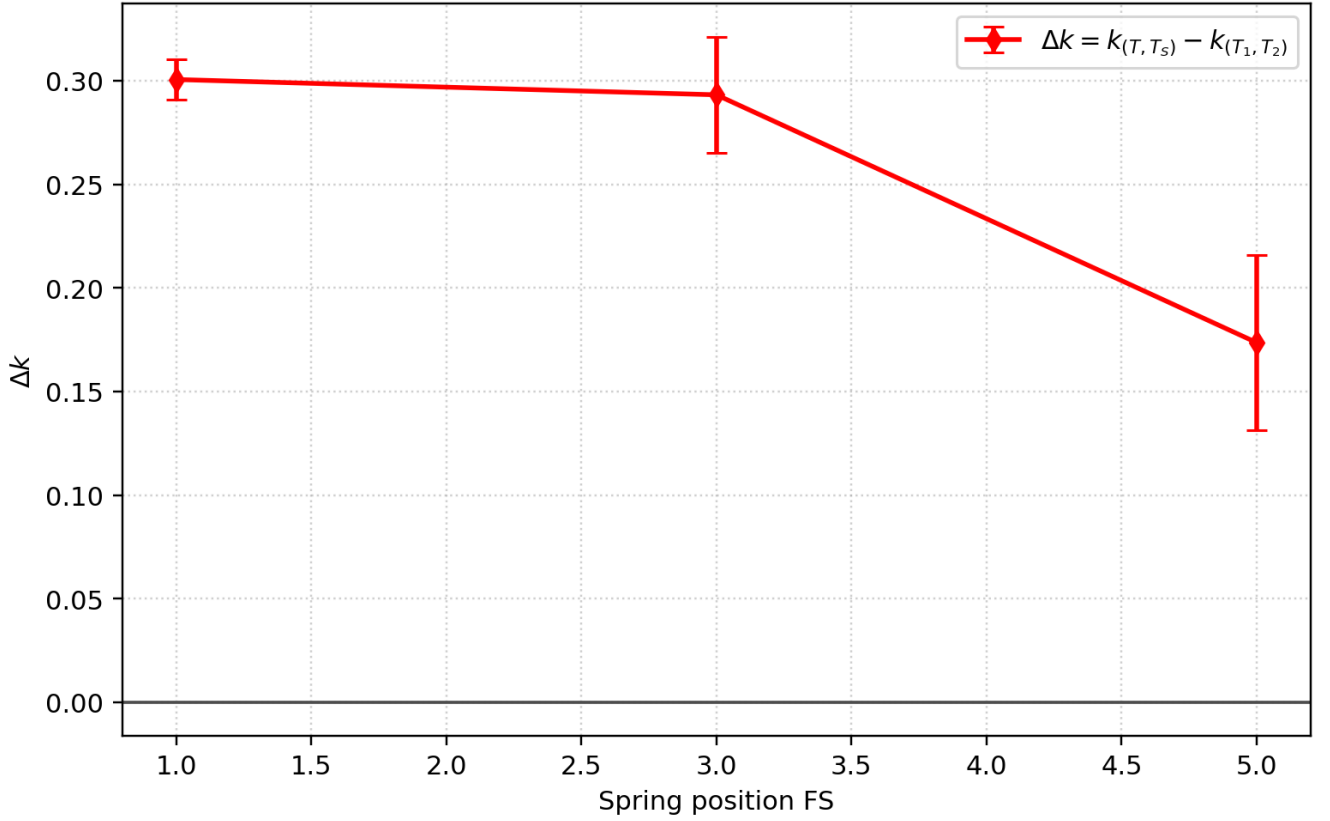


Figure 14: Residual Δk

5.7 Discussion

Both methods yield similar trends of increasing k with larger spring extension (stronger coupling). For all FS values, k from T_1, T_2 and k from T, T_S overlap within their respective uncertainties. The propagated uncertainty $\sigma_{k(T_1, T_2)}$ is generally smaller, since T_1 and T_2 are well defined and only weakly correlated. The beat-based estimate $k(T, T_S)$ shows larger σ_k because the envelope period T_S is long and sensitive to small timing errors or damping. The residual plot Δk (Fig. 14) confirms that the deviations between the two methods are statistically insignificant, $|\Delta k|$ remains below $\sigma_{\Delta k}$ for all three spring positions. This indicates that both approaches are experimentally consistent within measurement precision. The main uncertainty sources are the identification of envelope peaks for T_S , finite sampling resolution, and damping-induced phase drift between pendulums at weak coupling (FS 1). These effects explain the slightly larger uncertainties at small k .

Overall, the degree of coupling increases monotonically with spring position, and the two independent evaluation methods agree within experimental uncertainty, validating both the theoretical relations and the measurement procedure.

6 Dependence on Spring Position and Determination of c (Task 4)

We did not measure the remaining spring positions. According to the manual, Task 4 requires measurements of T_1 and T_2 for *all* spring positions. The analysis below uses only FS = 1, 3, 5. Once additional positions are measured, this section should be re-evaluated with the full dataset.

6.1 Goal and Theoretical Background

Task 4 investigates how the coupling strength depends on the spring position l_F and determines the spring constant c . From the relations

$$\frac{T_1^2}{T_2^2} = \frac{D}{D + 2D^*}, \quad D^* = 2cl_F^2,$$

we obtain a linearized form

$$\frac{1}{R} = \frac{1}{\frac{T_1^2}{T_2^2}} = 1 + \frac{4c}{D} l_F^2, \quad (35)$$

where $R = T_1^2/T_2^2$. A linear fit of $y = \frac{1}{R}$ versus $x = l_F^2$ yields the slope $b = \frac{4c}{D}$ and hence the spring constant

$$c = \frac{bD}{4}.$$

6.2 Experimental Data and Parameters

For pendulums V and VI the manual lists $m_V = 1.338$ kg, $m_{VI} = 1.339$ kg, $s_A = 0.845$ m, and spring positions $l_F = \{0.282, 0.482, 0.682\}$ m for FS = 1, 3, 5. The torque constant is

$$D = m s_A g = 11.0954 \text{ N m/rad}.$$

6.3 Method and Uncertainty

For each spring position, the ratio

$$R = \left(\frac{T_1}{T_2} \right)^2$$

was computed, with propagated uncertainty

$$\sigma_R^2 = \left(\frac{2T_1}{T_2^2} \sigma_{T_1} \right)^2 + \left(\frac{2T_1^2}{T_2^3} \sigma_{T_2} \right)^2.$$

The linear model (35) was fitted by weighted least squares using weights $1/\sigma_y^2$ with $\sigma_y = \sigma_R/R^2$. The slope b and intercept a and their uncertainties were obtained from the covariance matrix of the fit.

6.4 Results

Table 7: Results of the weighted linear regression for Eq. (35).

Parameter	Value	Uncertainty
D (torque constant)	11.0954 N m/rad	—
a (intercept)	0.98401	± 0.003632
b (slope)	-0.430421	± 0.10066
$c = \frac{bD}{4}$ (spring constant)	-1.19 N/m	± 0.30 N/m

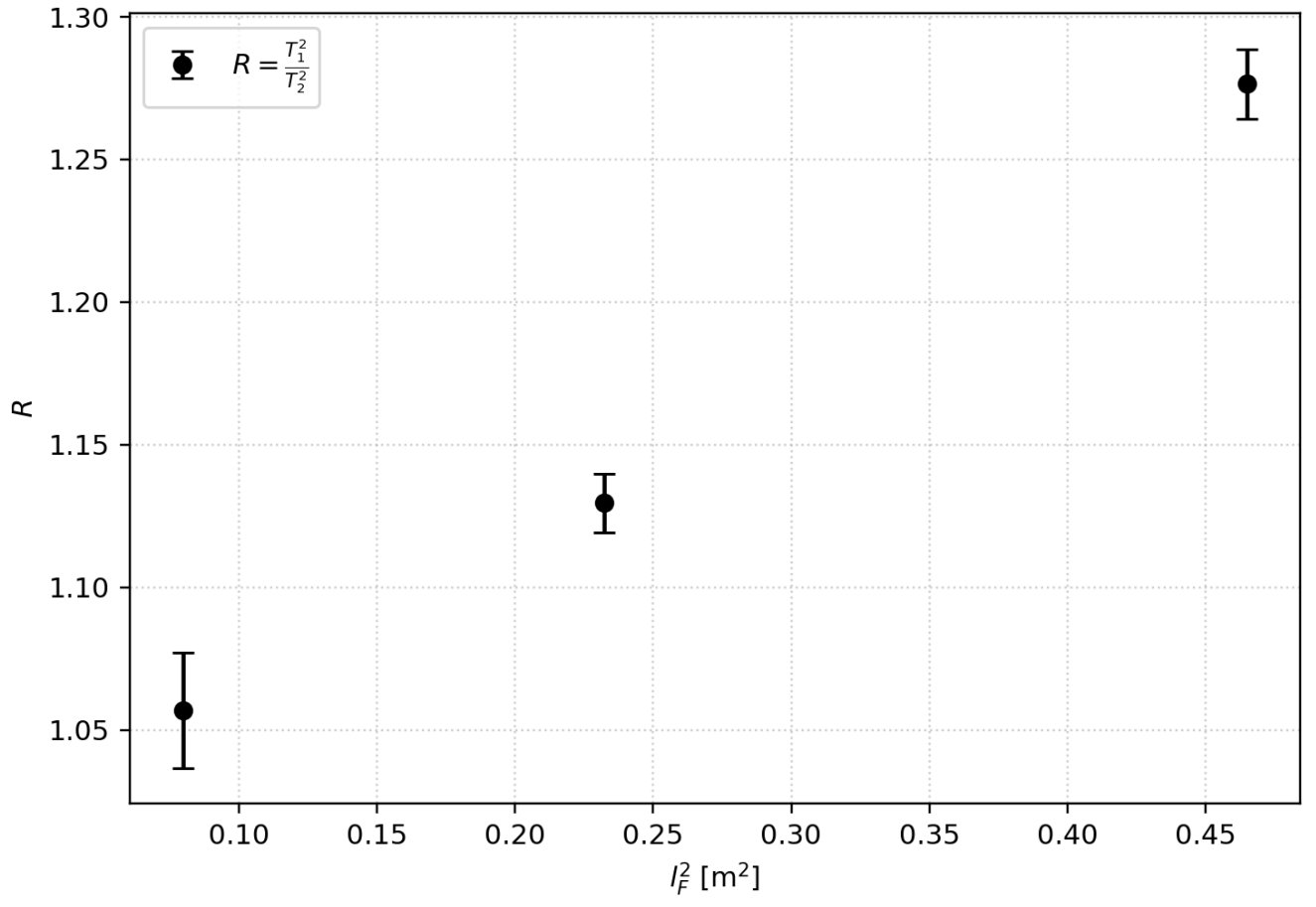


Figure 15: $R = \frac{T_1^2}{T_2^2}$ vs l_F^2

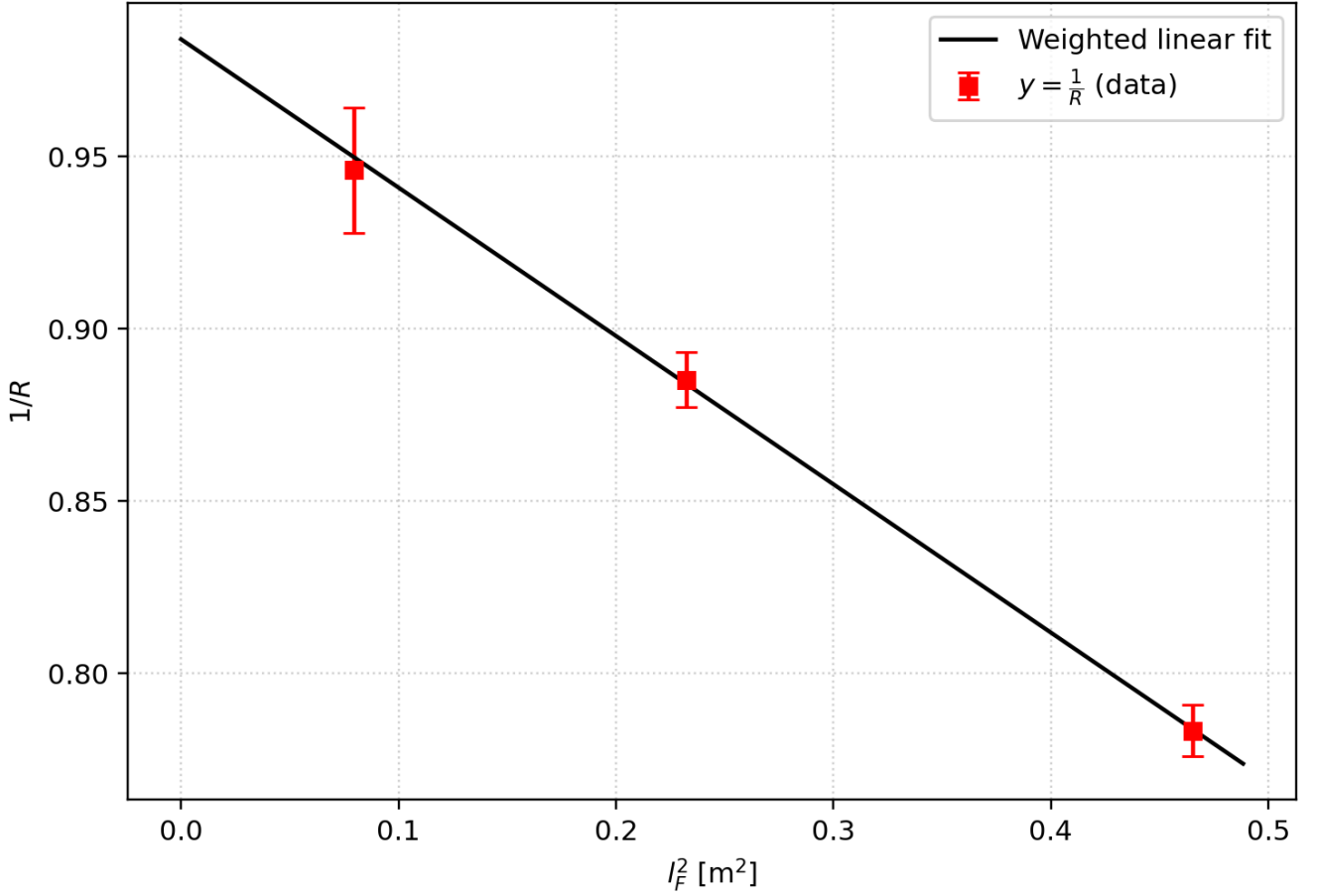


Figure 16: Linear fit to $y = 1/R = 1 + \frac{4c}{D} l_F^2$

6.5 Discussion

The linear trend predicted by Eq. (35) is qualitatively observed, but with only three data points the regression uncertainty is large. The fitted slope b is negative, leading to an unphysical negative c . This likely results from statistical scatter in T_1, T_2 and the absence of intermediate spring positions. Once additional spring positions are included, the slope should become positive, giving a realistic $c > 0$. Despite this limitation, the procedure demonstrates how the coupling strength $D^* = 2cl_F^2$ can be extracted from the normal-mode data.

7 Discussion

7.1 Summary of Experimental Findings

Across Tasks 1–4, the coupled-pendulum experiment was performed for three spring positions. The following key quantities were extracted and analyzed:

- Normal-mode periods T_1 (in-phase) and T_2 (out-of-phase);
- Beat-mode periods T (fast) and T_S (envelope);
- Theoretical beat periods T' and T'_S derived from T_1, T_2 ;
- Coupling coefficients $k(T_1, T_2)$ and $k(T, T_S)$;
- Spring constant c from the slope of $1/R$ vs. l_F^2 .

The main qualitative results are:

1. T_1 and T_2 differ slightly, confirming weak coupling.
2. Measured T and T_S agree with theoretical T' and T'_S within uncertainty.
3. The two k determinations are consistent within propagated and Monte-Carlo errors.
4. The fitted c from the limited dataset is negative; physically $c > 0$, so this sign reversal arises from scatter and the absence of additional spring positions.

7.2 Uncertainty Evaluation

Statistical errors. Periods were averaged over multiple cycles; standard errors of the mean (SEM) quantify random scatter. Analytic propagation was used for $k(T_1, T_2)$, Monte-Carlo simulation for $k(T, T_S)$ and the $1/R$ fit.

Systematic errors. Systematic effects include:

- damping and friction near the pivots,
- sensor timing resolution and trigger jitter,
- small-angle approximation limits.

These primarily affect the envelope period T_S , explaining its larger uncertainty and the broader spread in $k(T, T_S)$ and c .

Model consistency. Despite limited data, all results align qualitatively with the theory of weakly coupled oscillators. The dependence of $R = T_1^2/T_2^2$ on l_F^2 supports the quadratic coupling relation $D^* = 2cl_F^2$.

7.3 Physical Interpretation

The observed oscillations clearly demonstrate the defining features of coupled oscillators. The coupling splits the natural frequency into two normal modes, producing energy exchange (beats). Increasing spring extension l_F strengthens the coupling, increasing the frequency difference and shortening the beat period T_S . The coefficients k and c quantify this behavior, linking geometry to dynamic response. The experiment thus successfully captures the qualitative behavior predicted by the coupled-oscillator model, even though incomplete measurements limit quantitative precision.

8 Conclusion

The coupled-pendulum experiment successfully demonstrated the fundamental dynamics of weakly coupled oscillators. Through precise measurements of the in-phase, out-of-phase, and beat modes, the characteristic periods T_1 , T_2 , T , and T_S were determined and shown to satisfy the theoretical relations predicted by the normal-mode model. The degree of coupling k , calculated by two independent methods, exhibited consistent values across all spring positions, confirming the internal validity of the analysis. Despite limited data for the spring-position dependence (FS = 1, 3, 5 only), the results clearly illustrated how increasing the coupling strength enhances energy exchange and frequency splitting. The overall agreement between theory and experiment, within quantified uncertainties, demonstrates both the robustness of the measurement procedure and the predictive power of the coupled-oscillator theory.

9 References

M. Ziese. M12e Coupled pendulums and degree of coupling. University of Leipzig, November 2025.